GG-I-00521b(DSA-DM)
August 13, 1971
Interim Revision of
GG-I-00521a(DSA-DM)
August 25, 1967

# INTERIM FEDERAL SPECIFICATION

# INDICATOR, STERILIZATION

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-I-521.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of sterilization indicators, suitable for use in determining whether or not sterilizing temperature has been reached in the sterilizing chamber of autoclaves and sterilizers.

1.2 Classification.

1.2.1 Types. The Indicators, Sterilization, covered by this specification shall be of the following types:

| Type | Fusible Temperature, degrees Fahrenheit | Color change |
|---|---|---|
| I | 248° + 4°. - 2° | Off-white to Red Brown |
| II | 272° + 4°. - 2° | White to Rose Pink or Light blue to dark blue |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |
| PPP-B-676 | Box, Setup. |

Military

| | |
|---|---|
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible. |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-191 | Textile Test Methods. |
| FED-STD-123 | Marking for Domestic Shipment (Civilian Agencies). |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, DC.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbocks and the Index of Federal Specifications, Standards, and Handbocks from established distribution points in their agencies.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

3.1.1 Glass. Indicator tube shall be clear and colorless soda lime glass.

* 3.1.2 Thread. Indicator thread shall be linen or cotton. Thread shall have breaking strength of not less than four pounds when tested as specified in 4.4.3. The dyed thread shall not bleed when tested as specified in 4.4.2.

* 3.1.3 Indicator tablets. Indicator tablets shall be as hereinafter specified. The tablets shall be solid material whose melting points shall be, 248° + 4°, -2° F., for Type I, and 272° + 4° - 2° F. for Type II when tested as specified in 4.4.1. The tablet color shall change from off-white to red brown for Type I and from white to rose pink or from light blue to dark blue for Type II. The rose pink color shall be at least as dark as the shade of the rose pink of the thread tied to the tube.

3.2 Construction. Sterilization indicator shall consist of a straight, heat sealed, glass tube, 44.0 ± 4.0 millimeters in length and 5.5 ± 0.5 millimeters in diameter. The tube shall contain the indicator tablet. The tablet shall melt and a positive change in color or shade, as applicable to the type, shall occur within the temperature range specified. At one end of the tube, there shall be a constriction to which shall be secured a length of linen or cotton thread. The thread shall be of sufficient length so that when tied to the glass tube, there are two lengths of thread each not less than 305 millimeters long.

3.3 Instructions. Full instructions for use shall accompany each unit of 100 indicators.

3.4 Workmanship. Workmanship shall be first class throughout. The finished sterilization indicators shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor, and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conductd in accordance with MIL-STD-105 and Table I.

Table I. Sampling for Examination*

| | Inspection level | AQL (defects per hundred units) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 1.0 |
| Minor defects | II | 4.0 |
| For dimensional examination | S-3 | 2.5 |

*Unit for sampling shall be one tube.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II.

Table II. Sampling for Tests

| Component | Characteristic | Requirement | Test procedure | Inspection level | AQL |
|---|---|---|---|---|---|
| Individual Indicator | Melting point | 3.1.3 | 4.4.1 | S-2 | * |
| Thread | Colorfast | 3.1.2 | 4.4.2 | 1 Yard per lot of thread | 4.0 |
| Thread | Breaking strength | 3.1.2 | 4.4.3 | Ten 16" lengths per lot of thread | 2.5 |

*Acceptance number shall be zero, and not less than five samples shall be taken at random from each unit regardless of lot size. The sample for this test shall be the individual indicator tube.

4.3 Examination. The indicator shall be examined to determine conformance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects shown in Table III.



Source: https://assist.dla.mil -- Downloaded: 202610-26T11:51Z
Check the source to verify that this is the current version before use.

Table III. Classification of defects*

| Categories | Defects |
|---|---|
| Major | |
| 101 | Glass tube not free of cracks and defective heat seals. |
| 102 | Glass tube not free of imperfections which will interfere with the observation of the indicator tablet. |
| 103 | Indicator tablet not included in tube. |
| Minor | |
| 201 | Indicator tube insufficiently constricted at one end to retain thread. |
| 202 | Thread insecurely fastened to tube. |
| 203 | Indicator tablet not in single pellet form. |
| 204 | Full instructions for use not included. |

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The indicators shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted as necessary to determine compliance with specification requirements.

4.4.1 Determination of melting point. Melting point of the indicator shall be determined by attaching the indicator tube to a thermometer calibrated to 250° F. ± 1° F. for Type I indicators, and 270° F. ± 1° F. for Type II indicators. The indicator tube shall be adjacent to the thermometer bulb. The thermometer, with indicator tube attached, shall be placed in an oil bath (preheated to 212° F.) so that the indicator tube is completely immersed in the oil. The temperature of the oil shall be raised slowly until 218° F. is reached for Type I indicators, and 240° F. for Type II indicators. Heating shall continue, with constant stirring, so that the temperature rises at a rate of 3° F. per minute. When the temperature reaches 248° F. for Type I indicators and 267° for Type II indicators, heating shall be reduced so that the temperature rises at a rate of 1° F. to 2° F. per minute. Heating shall continue until melting is complete. The temperature at which the tablet is observed to collapse definitely against the side of the tube shall be recorded as the beginning of melting, and the temperature at which the tablet becomes liquid throughout shall be recorded as the end of melting or the melting point. The sample shall be removed from the bath, allowed to cool and re-solidify. Sample shall then be examined for fused condition and color change.

4.4.2 Colorfastness of thread. A one-yard length of dyed thread shall be coiled and placed between layers of white gauze or cotton sheeting thoroughly moistened with distilled water and wrapped in dry huck towel and autoclaved for 20 minutes at 250° F., 15 pounds per square inch, and dried.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

4.4.3 Breaking strength. Breaking strength shall be determined in accordance with method 4100 of FED-STD-191.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One box containing 100 indicators, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 12 units | 144 units |

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676. Box design shall include partitions or die cut platform, as indicated in figure 12 or 13 of PPP-B-676, to hold each indicator in an individual cell. Closure of box shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, grade V3C.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 For civil agencies. Marking shall be in accordance with Fed-Std No. 123. Lot numbers shall be printed on unit packages.

5.3.2 For military agencies.

5.3.2.1 Interior package. Each unit package and each intermediate package shall be marked in accordance with Military Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name of contractor shall be shown. Date of manufacture shall be shown in lieu of date packed.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

5.3.2.2 Exterior containers. Exterior container shall be marked in accordance with Military Standard MIL-STD-129. Lot (control) number and date of manufacture in lieu of date packed shall be shown.

6. NOTES

* 6.1 Intended use. For use in determining whether or not sterilizing temperature has been reached in the sterilizing chamber of autoclaves and sterilizers.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).
(c) Whether Type I or Type II is required.

* 6.3 This specification covers the following items appearing in the Federal Supply Catalog:

| FSN | Item Identification | Specification Designation Type |
|---|---|---|
| 6530-708-5320 | INDICATOR, STERILIZATION, TABLET in Tube, 248° + 4° - 2° F, 100s | I |
| 6530-935-1124 | INDICATOR, STERILIZATION, TABLET in Tube, 272° + 4° - 2° F, 100s | II |

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6530-1231

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.